

PROJECT ESTIMATION & DEVELOPMENT PHASE PLAN

Project: AI-Based Real-Time Live Speech Translation System

Scope: Hindi to Telugu Real-Time Speech-to-Speech Translation Platform

Platform: Web-Based Real-Time Translation Infrastructure

PROJECT OVERVIEW

The proposed system includes:

- Real-time speech recognition
- Live language translation
- AI-based speech synthesis output
- Real-time streaming architecture
- Event audio integration
- Web-based control dashboard
- Multi-user/event management capability
- Cloud-based AI integrations and processing

The system is planned as an MVP (Minimum Viable Product) within approximately 2 months of development and testing.

PHASE 1 — INITIAL DEVELOPMENT & FOUNDATION

Duration: Week 1 – Week 2

Scope:

- Frontend architecture setup
- Backend server setup
- Real-time audio streaming setup
- WebSocket integration
- Initial Azure/OpenAI API integration
- Project infrastructure configuration
- Database and cloud environment setup
- Basic dashboard UI implementation
- Audio input/output testing setup

Estimated Cost:

₹30,000 – ₹40,000

Payment Purpose:

Initial development investment, infrastructure setup, API onboarding, realtime streaming architecture, and foundational implementation.

PHASE 2 — AI INTEGRATION & LIVE PROCESSING

Duration: Week 3 – Week 4

Scope:

- Azure Speech Recognition integration
- Live translation processing pipeline
- OpenAI contextual translation enhancement
- ElevenLabs voice synthesis integration
- Real-time audio processing optimization
- Telugu speech output implementation
- Latency optimization and buffering
- Internal testing and debugging

Estimated Cost:

₹30,000+

Payment Purpose:

Advanced AI integration, live translation processing, voice generation implementation, and streaming optimization/testing.

PHASE 3 — TESTING, OPTIMIZATION & FINAL MVP

Duration: Week 5 – Week 8

Scope:

- Real-event simulation testing
- Speaker/mixer integration testing

- Stability optimization
- Cloud deployment configuration
- Performance tuning
- Multi-session handling
- Dashboard improvements
- Final MVP preparation
- Production-level testing and refinements

Estimated Cost:

Remaining project balance based on final implementation scope and testing requirements.

ESTIMATED PROJECT RANGE

Estimated MVP Development Cost:

₹1.25 Lakhs+ (Approx.)

Final project costing may vary depending on:

- Additional language requirements
- Voice customization requirements
- Advanced AI processing requirements
- Deployment scale
- Event concurrency requirements
- Infrastructure scaling and cloud usage

ESTIMATED AI SERVICES & INFRASTRUCTURE COST

Service / Infrastructure	Purpose	Estimated Monthly Cost
Azure Speech Services	Speech Recognition & Translation	₹4,000 – ₹10,000
OpenAI API Usage	Contextual AI Translation Processing	₹3,000 – ₹8,000
ElevenLabs Voice AI	AI Voice Synthesis Output	₹2,000 – ₹7,000
Cloud Hosting & Streaming	Backend Hosting & Realtime Streaming	₹2,000 – ₹5,000
Database & Monitoring	Storage, Logs & Monitoring	₹1,000 – ₹3,000

Estimated Operational Range:

₹10,000 – ₹30,000/month

(Depending on usage scale, event traffic, and AI processing volume)

PAYMENT STRUCTURE & DEVELOPMENT MILESTONES

Payment Stage	Timeline	Estimated Amount	Purpose
Initial Payment	Project Start (Week 1)	₹30,000 – ₹40,000	Frontend/backend setup, infrastructure configuration, API onboarding, realtime streaming foundation
Second Payment	After 2 Weeks (Week 3–4)	₹30,000+	AI integrations, translation pipeline, voice synthesis setup, optimization & testing
Final Payment	Before Final Delivery / MVP Completion	Remaining Balance	Final testing, event optimization, deployment preparation, refinements & production readiness

NOTE

This estimation is prepared for MVP-level development and initial deployment architecture. Further enterprise-scale enhancements, additional language support, offline processing, mobile applications, or advanced voice cloning systems may require separate scaling and infrastructure planning.